

OZ — The Ozone Story

OCR B (Salters) H433 storyline **OZ**. Ozone as a sunscreen: radiation and matter, rates of reaction and catalysis, radical reactions, intermolecular bonding, and the haloalkanes of CFCs and HFCs with their nucleophilic substitution.

1. Rates of Reaction and Collision Theory

Collision Theory

Collision theory: For a reaction to occur, particles must collide with sufficient energy (equal to or greater than the activation energy, E_a) and with the correct orientation.

Activation energy (E_a): The minimum energy that colliding particles must have in order to react. It represents the energy barrier that must be overcome for bonds to break and new bonds to form.

Factors Affecting Rate

Factor	Effect	Explanation
Concentration (or pressure for gases)	Increases rate	More particles per unit volume → more frequent collisions → more successful collisions per unit time
Temperature	Increases rate	Particles have greater kinetic energy → more frequent collisions AND a greater proportion of particles exceed E_a (the latter is the dominant effect)
Surface area	Increases rate	More surface exposed to reactant particles → more frequent collisions

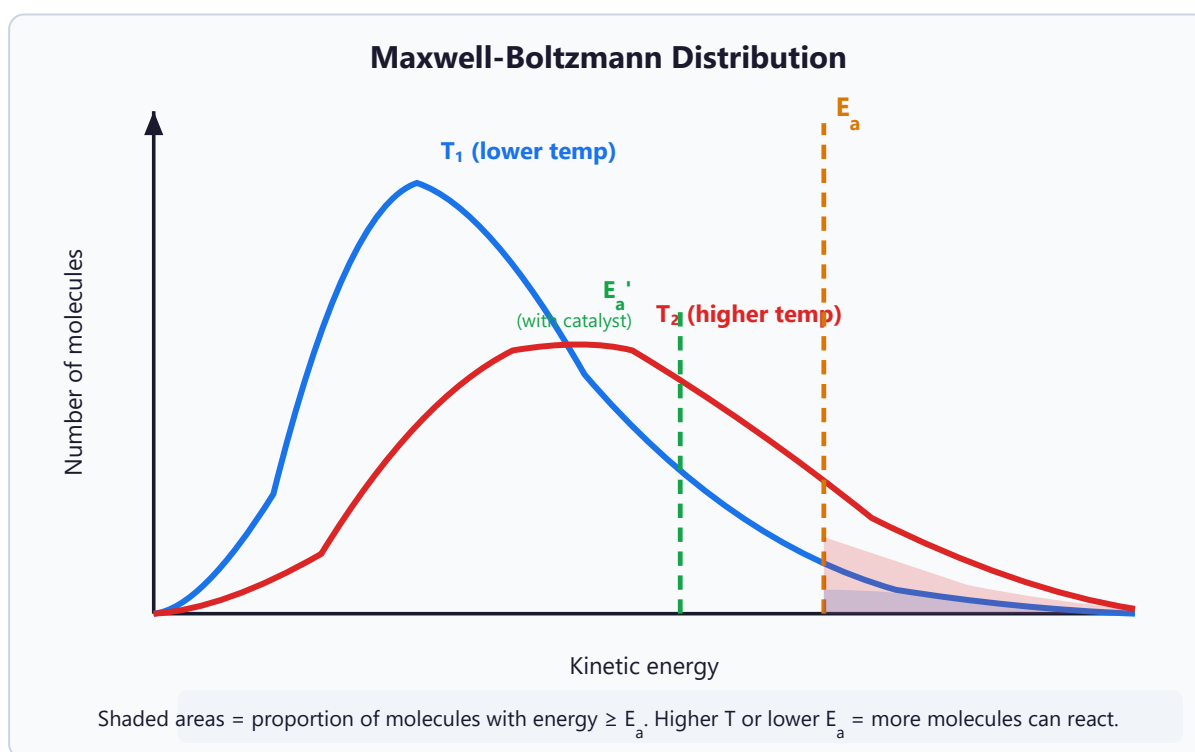
Catalyst	Increases rate	Provides an alternative reaction pathway with a lower activation energy
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The Boltzmann Distribution

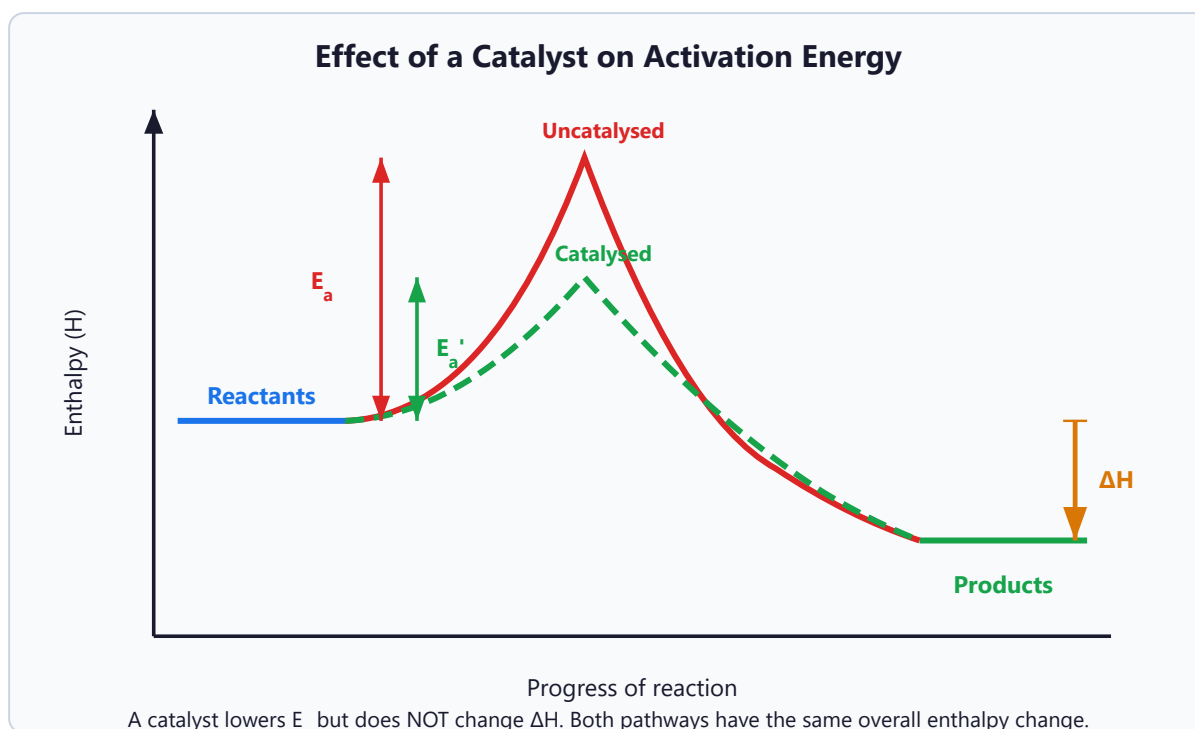
The Boltzmann (Maxwell-Boltzmann) distribution shows the distribution of molecular kinetic energies in a sample of gas at a given temperature. Key features:

- No molecules have zero energy.
- The distribution is asymmetric with a long tail to the right (some molecules have very high energies).
- The peak (most probable energy) shifts to the right and becomes lower and broader at higher temperatures.
- The area under the curve is constant (total number of molecules is unchanged).
- The shaded area to the right of E_a represents the proportion of molecules with enough energy to react.

Increasing temperature shifts the distribution to the right and significantly increases the proportion of molecules exceeding E_a . A catalyst lowers E_a , so a greater proportion of molecules exceed the new, lower E_a without changing the distribution itself.



Catalysts



Catalyst: A substance that increases the rate of a reaction by providing an alternative pathway with a lower activation energy, without being consumed in the overall reaction.

Homogeneous catalyst: A catalyst in the same phase as the reactants (e.g. $\text{H}^+(\text{aq})$ catalysing an esterification in aqueous solution).

Heterogeneous catalyst: A catalyst in a different phase from the reactants (e.g. iron in the Haber process, vanadium(V) oxide in the Contact process). Reactions occur on the surface of the catalyst, so a greater surface area increases effectiveness.

2. Halogenoalkanes

Halogenoalkane: An organic compound in which one or more hydrogen atoms of an alkane have been replaced by halogen atoms (F, Cl, Br, I). The C–X bond is polar because halogens are more electronegative than carbon, making the carbon atom susceptible to attack by nucleophiles.

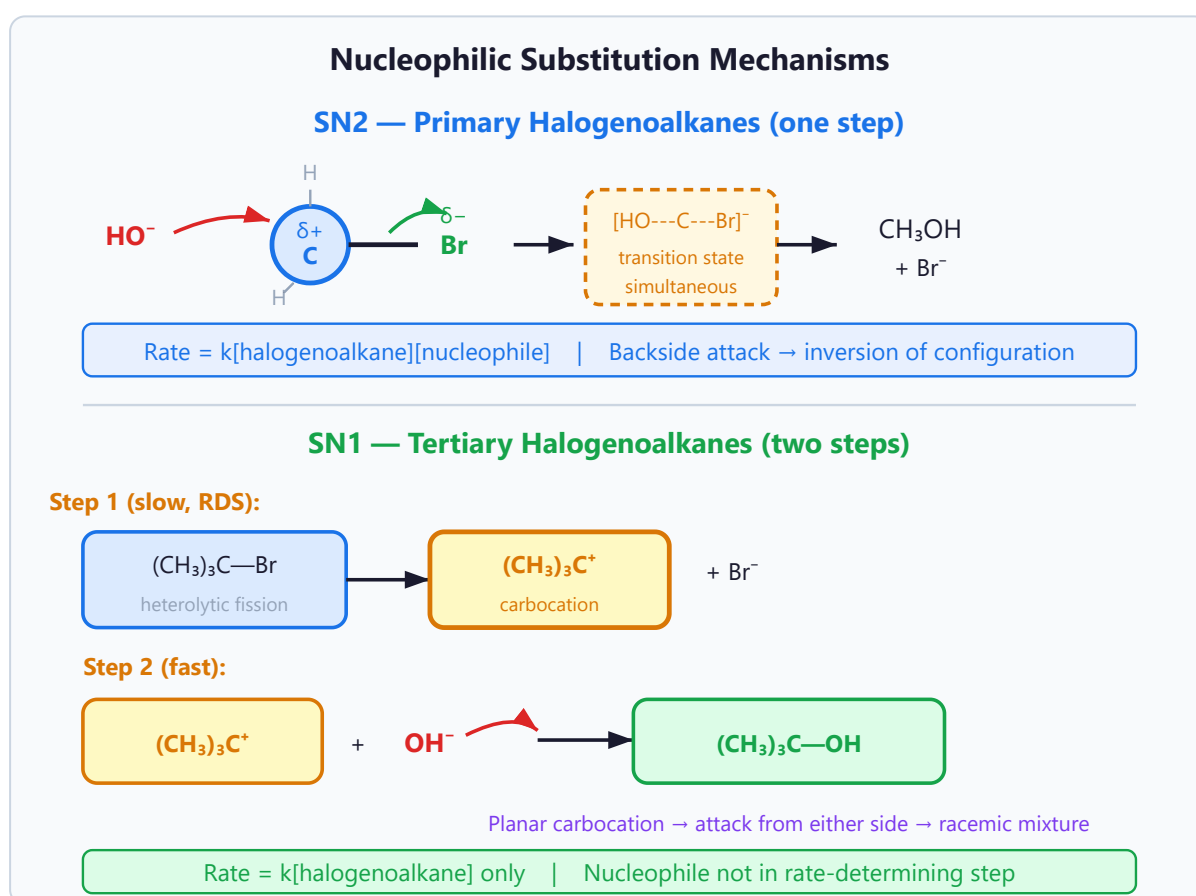
Nucleophilic Substitution

Nucleophile: An electron-pair donor; a species with a lone pair that it can donate to an electron-deficient atom. Examples: OH^- , CN^- , NH_3 , H_2O .

SN2 mechanism (primary halogenoalkanes): A one-step mechanism. The nucleophile attacks the δ^+ carbon from the opposite side to the halogen (backside attack). Bond making and bond breaking happen simultaneously in a single transition state. The rate equation is: $\text{rate} = k[\text{halogenoalkane}][\text{nucleophile}]$. SN2 produces inversion of configuration at the chiral centre.

SN1 mechanism (tertiary halogenoalkanes): A two-step mechanism. Step 1: the C–X bond breaks heterolytically to form a carbocation and a halide ion (slow, rate-determining step). Step 2: the nucleophile rapidly attacks the planar carbocation from either side. The rate equation is: $\text{rate} = k[\text{halogenoalkane}]$. SN1 produces a racemic mixture if the carbon was a chiral centre, because the planar carbocation can be attacked equally from both sides.

Secondary halogenoalkanes undergo a mixture of SN1 and SN2 mechanisms.

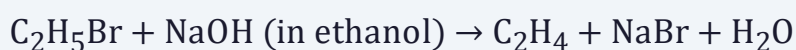


Reactions of Halogenoalkanes

Nucleophile	Conditions	Product
NaOH(aq) / OH ⁻	Warm aqueous solution	Alcohol
KCN / CN ⁻	Warm ethanol/water	Nitrile (extends carbon chain by 1)
NH ₃	Excess, warm, sealed tube	Amine

Elimination

When a halogenoalkane is heated with a strong base in ethanolic solution (e.g. NaOH in ethanol), elimination occurs instead of substitution. A hydrogen atom and the halogen are removed from adjacent carbons, forming an alkene and water:



Exam Tip: The choice between substitution and elimination depends on the conditions. Aqueous NaOH favours substitution (nucleophilic); ethanolic NaOH favours elimination (OH⁻ acts as a base). Tertiary halogenoalkanes favour elimination and S_N1; primary halogenoalkanes favour S_N2.

Worked Example — Percentage Yield:

A student reacted 5.00 g of butan-1-ol with excess ethanoic acid. The theoretical yield of the ester (butyl ethanoate, M_r = 116.2) is 7.85 g. The actual yield was 4.50 g.

$$\text{Percentage yield} = (4.50 / 7.85) \times 100 = \mathbf{57.3\%}$$

The yield is below 100% due to: the reversible nature of esterification, loss during transfer and separation, incomplete reaction, and evaporation of volatile ester.

Reactivity of Halogenoalkanes

The rate of hydrolysis (reaction with aqueous silver nitrate to test for the halide ion) decreases in the order: R-I > R-Br > R-Cl. This is because the C-I bond is the weakest (lowest bond enthalpy) and easiest to

break, despite being the least polar. Bond strength is the dominant factor, not polarity.

3. Organic Mechanisms Summary

Curly arrow: Represents the movement of a pair of electrons. A curly arrow starts from an electron-rich source (lone pair or bond) and points to an electron-deficient destination. A single-headed (fishhook) arrow represents the movement of a single electron (used in radical mechanisms).

Types of organic reaction:

- **Addition:** Two molecules combine to form one product (atoms add across a double bond).
- **Substitution:** An atom or group is replaced by another atom or group.
- **Elimination:** A small molecule (e.g. H_2O , HBr) is removed from a larger molecule, forming a double bond.
- **Oxidation/Reduction:** Gain or loss of oxygen/hydrogen, or change in oxidation state.
- **Hydrolysis:** A bond is broken by the addition of water (or OH^-).

Exam Tip: When drawing mechanisms, every curly arrow must start from a lone pair or bond and point to where the electrons are going. Show relevant partial charges ($\delta+$ and $\delta-$) on polar bonds. For ionic intermediates, show the charge clearly. Marks are awarded for correct curly arrows, intermediate species, and products.

4. The Atmosphere and the Ozone Layer

Composition of the Atmosphere

Gas mixtures are quoted by **volume**, not by mass, because equal volumes of gases at the same temperature and pressure contain equal numbers of molecules. A percentage by volume is therefore also a percentage by moles.

Gas	% by volume (dry air)	Parts per million (ppm)
Nitrogen, N_2	78.1	781 000

Oxygen, O ₂	20.9	209 000
Argon, Ar	0.93	9 300
Carbon dioxide, CO ₂	about 0.043 (2026)	about 430
Ozone, O ₃ (stratosphere)	about 0.001 at most	up to about 10

Parts per million (ppm): The number of volumes of a component in every 10⁶ volumes of the mixture. To convert, percentage $\times 10^4 = \text{ppm}$, and $\text{ppm} \div 10^4 = \text{percentage}$.

Worked Example — Composition by Volume:

A sample of urban air is found to contain nitrogen dioxide at 0.000 008% by volume. Express this in ppm, and calculate the volume of NO₂ in 50.0 dm³ of this air.

$$\text{ppm} = 0.000\,008 \times 10^4 = \mathbf{0.08\,ppm}$$

$$\text{Volume of NO}_2 = 50.0 \times (0.08 / 10^6) = 4.0 \times 10^{-6} \text{ dm}^3 = \mathbf{4.0 \times 10^{-3} \text{ cm}^3}$$

Notice how small the numbers are. Ozone and the oxides of nitrogen matter enormously at concentrations of a fraction of a ppm, which is exactly why ppm is the useful unit for atmospheric chemistry.

All the ozone in the atmosphere, brought down to the surface and compressed, would form a layer only about 3 mm thick. It is spread through the stratosphere, roughly 15–30 km up.

Radiation and Bond Breaking

In order of increasing frequency and energy (and decreasing wavelength): **infrared** → **visible** → **ultraviolet**. The Sun radiates strongly across all three; the cooler Earth radiates mainly in the infrared.

$$E = hf \quad \text{and} \quad f = c / \lambda \quad \text{so} \quad E = hc / \lambda$$

where $h = 6.63 \times 10^{-34} \text{ J Hz}^{-1}$ and $c = 3.00 \times 10^8 \text{ m s}^{-1}$. Each of the three regions does something different to a molecule:

Region	Effect on a molecule
Infrared	Makes bonds vibrate (stretch and bend) more — the basis of IR spectroscopy and of the greenhouse effect
Visible and ultraviolet	Promotes electrons to higher energy levels; if enough energy is absorbed, a bond can break

Worked Example — Can This Radiation Break the Bond?

The O=O bond enthalpy in oxygen is 498 kJ mol^{-1} . Show that ultraviolet radiation of wavelength 240 nm can dissociate an oxygen molecule.

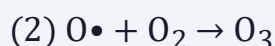
$$E = hc / \lambda = (6.63 \times 10^{-34} \times 3.00 \times 10^8) / (240 \times 10^{-9}) = 8.29 \times 10^{-19} \text{ J per photon}$$

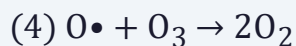
$$\text{Per mole: } 8.29 \times 10^{-19} \times 6.02 \times 10^{23} = 4.99 \times 10^5 \text{ J mol}^{-1} = \mathbf{499 \text{ kJ mol}^{-1}}$$

499 kJ mol^{-1} is just greater than the 498 kJ mol^{-1} needed, so radiation of 240 nm (and anything shorter) breaks the O=O bond. Radiation of longer wavelength — say 320 nm, worth only 374 kJ mol^{-1} — cannot.

How Ozone is Made and Destroyed

Ozone in the stratosphere is in a steady state: four reactions, running continuously, keep its concentration roughly constant.





Reactions (1) and (2) make ozone; reactions (3) and (4) destroy it. The crucial point is that the ozone layer works as a sunscreen because of steps (1) and (3) *together*: oxygen absorbs the very shortest, most energetic UV, and ozone absorbs the longer-wavelength UV that oxygen lets through. Nothing else in the atmosphere absorbs in that window.

UV and living things: UV-C (< 280 nm) is absorbed almost entirely by O_2 and O_3 . UV-B (280–315 nm) is absorbed mainly by ozone, and what gets through causes sunburn, damages DNA and raises the risk of skin cancer and cataracts. UV-A (315–400 nm) passes through largely unabsorbed and causes skin ageing.

Ozone in the Troposphere is a Pollutant

Ozone is only useful where it is. At ground level it is a toxic irritant that inflames the lungs, aggravates asthma, damages crops and attacks rubber and plastics. It is a secondary pollutant: it is not emitted directly but forms in sunlight from nitrogen oxides and unburnt hydrocarbons released by vehicles.



The resulting mixture of ozone, NO_2 and organic products is **photochemical smog**, worst on hot, still, sunny days in traffic-heavy cities.

Exam Tip: "Good up high, bad nearby" is the one-line summary, but an exam answer needs the reason: stratospheric ozone screens out UV-B that would otherwise reach the surface, while tropospheric ozone is a lung irritant at the concentrations found in smog. The *chemistry* of the molecule is identical in both places; only its location changes whether it helps or harms.

5. Radical Reactions and the CFC Story

Homolytic and Heterolytic Fission

	Homolytic fission	Heterolytic fission

How the pair splits	One electron to each atom	Both electrons to one atom
Products	Two radicals: $X\cdot$ and $Y\cdot$	Two ions: X^+ and Y^-
Arrow used	Half curly arrow (single-headed, "fishhook") — one electron moves	Full curly arrow (double-headed) — a pair of electrons moves
Typical setting	Non-polar bonds, gas phase, UV light or high temperature	Polar bonds, solution

Radical: A species with an unpaired electron, shown by a dot, e.g. $Cl\cdot$ or $\cdot CH_3$. Radicals are extremely reactive because the unpaired electron readily pairs up, and they are usually very short-lived.

Radical Chain Reactions

A radical chain reaction has three stages. The classic example is the reaction of methane with chlorine in UV light:

Stage	What it does	Equations
Initiation	UV light breaks a bond homolytically to make the first radicals. Radicals are created.	$Cl_2 \rightarrow 2Cl\cdot$
Propagation	A radical reacts to give a product <i>and a new radical</i> , which keeps the chain going. The number of radicals is unchanged.	$Cl\cdot + CH_4 \rightarrow \cdot CH_3 + HCl$ $\cdot CH_3 + Cl_2 \rightarrow CH_3Cl + Cl\cdot$
Termination	Two radicals combine. Radicals are destroyed and that chain stops.	$Cl\cdot + Cl\cdot \rightarrow Cl_2$ $\cdot CH_3 + Cl\cdot \rightarrow CH_3Cl$ $\cdot CH_3 + \cdot CH_3 \rightarrow C_2H_6$

Adding the two propagation steps gives the overall reaction, $\text{CH}_4 + \text{Cl}_2 \rightarrow \text{CH}_3\text{Cl} + \text{HCl}$. Because one initiation event can drive thousands of propagation cycles, a tiny amount of light produces a large amount of product.

The reaction is a poor synthetic method, because the chloromethane produced is itself attacked by chlorine radicals to give CH_2Cl_2 , CHCl_3 and CCl_4 , and because the ethane from termination contaminates the product. That is exactly the kind of mixture green chemistry tries to avoid.

Exam Tip: The propagation test is simple and marks hang on it: a radical must appear on *both* sides of a propagation step. If radicals only appear on the right it is initiation; if they only appear on the left it is termination. Use half curly arrows (single-headed) in radical mechanisms and full curly arrows in ionic mechanisms — mixing the two loses marks.

CFCs: the Right Properties for the Wrong Reason

Chlorofluorocarbons such as CCl_3F (CFC-11) and CCl_2F_2 (CFC-12) were adopted in the 1930s as refrigerants, aerosol propellants, foam-blowing agents and solvents. They were chosen because they are non-toxic, non-flammable, volatile and, above all, chemically inert.

That last property is the problem. Being inert, CFC molecules are not broken down by anything in the troposphere — they are not attacked by water, oxygen or hydroxyl radicals — so they survive for decades and drift slowly up into the stratosphere. There, for the first time, they meet UV of short enough wavelength to break a bond.

Which Bond Breaks?

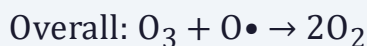
Bond	Bond enthalpy / kJ mol^{-1}	Longest wavelength able to break it / nm
C–F	467	256
C–Cl	346	346
C–Br	290	413
C–I	228	525

Bond enthalpy falls down the group as the C–X bond gets longer, so photodissociation gets easier from fluoro- to iodo-compounds. In the stratosphere the C–Cl bond is the weak link: the UV present there has plenty of energy to break C–Cl but not enough to break the much stronger C–F bond. So a CFC releases chlorine radicals, not fluorine radicals:

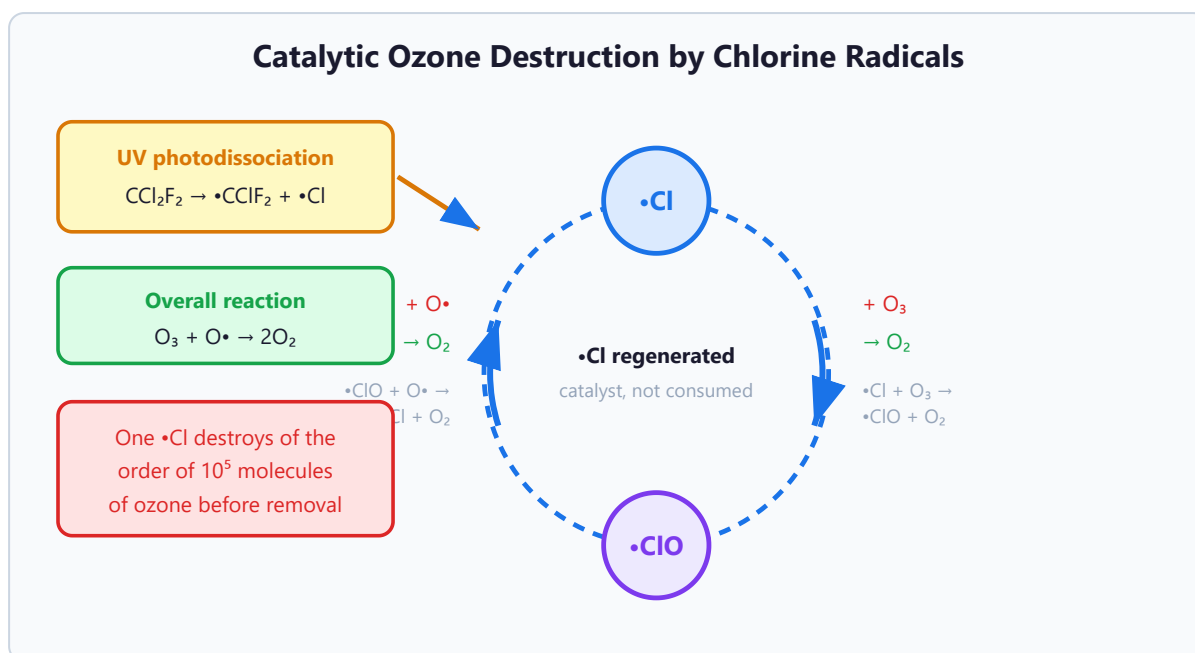


Catalytic Destruction of Ozone

A single chlorine radical then runs a catalytic cycle:



The chlorine radical is regenerated in the second step, so it is not consumed: it is a **homogeneous catalyst** (same phase as the reactants) and $\bullet\text{ClO}$ is the reaction intermediate. It provides an alternative route of lower activation enthalpy for a reaction that would otherwise be slow. One chlorine radical can destroy of the order of 10^5 ozone molecules before it is finally removed, which is why parts per *trillion* of CFCs were able to thin the whole ozone layer.



The Antarctic Ozone Hole

Ozone loss is far more severe over Antarctica than anywhere else, and three features of the polar winter explain why. The circumpolar vortex isolates the air over the continent; temperatures fall low enough to form polar stratospheric clouds of ice; and reactions on the surfaces of those ice particles convert the chlorine held in reservoir molecules such as HCl and ClONO_2 into Cl_2 . When the Sun returns in the Antarctic spring, that stored Cl_2 is photodissociated all at once, and ozone is destroyed very rapidly.

Replacing CFCs

Replacement	Structure	Why it is better	Remaining problem
HCFCs	Contain C–H as well as C–Cl, e.g. CHClF_2	The C–H bond is attacked by hydroxyl radicals in the troposphere, so most molecules are destroyed before reaching the stratosphere	Still contain chlorine, so a small ozone-depleting potential remains; now being phased out
HFCs	Contain only C, H and F, e.g. CH_2FCF_3	No chlorine at all, so essentially zero ozone-depleting potential	Powerful greenhouse gases, which is why they are themselves being phased down

Hydrocarbons and CO ₂	e.g. propane, butane, CO ₂	No chlorine and very low global-warming potential	Hydrocarbons are flammable; CO ₂ systems run at high pressure
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The international response is the **Montreal Protocol**, agreed in 1987 and since ratified by every UN member state, which phased out CFC production. The 2016 **Kigali Amendment** extended it to phase down HFCs on climate grounds. Atmospheric chlorine levels peaked in the late 1990s and have been falling since. The 2022 WMO/UNEP scientific assessment — the most recent full assessment as of 2026 — projected that, if current controls hold, ozone will return to 1980 values by about 2040 over most of the world, about 2045 over the Arctic and about 2066 over the Antarctic.

Exam Tip: A very common exam question asks why CFCs were used *and* why they are a problem, and the answer to both is the same word: unreactive. Being unreactive made them safe to handle and gave them a long enough atmospheric lifetime to reach the stratosphere intact. Make that link explicitly rather than listing the two facts separately.